Papaloizou–Pringle torus in Phantom (SPH)

Formulary

1 Units: from the Athena++ torus to $G = M = R_0 = 1$

Athena++ (athena-collab): $L_0 = \chi r_g$, $v_0 = c_{s0}$, $t_0 = L_0/c_{s0}$, $r_g = 2GM/c^2$, $c_{s0}^2 = \gamma k_B T_0/(\mu m_p)$, gravity $\tilde{\Phi} = -\beta/\tilde{r}$, $\beta = c^2/(2\chi c_{s0}^2)$. Phantom: same L_0, mass unit $M = M_{\rm bh}$, $G = 1$:

$$GM = \frac{c^2 r_g}{2} = \frac{c^2 L_0}{2\chi}, \qquad t_u = \sqrt{\frac{L_0^3}{GM}} = \frac{L_0\sqrt{2\chi}}{c}, \qquad \frac{t_0}{t_u} = \frac{c}{c_{s0}\sqrt{2\chi}} = \sqrt{\beta}.$$

$$t_{\rm ph} = \sqrt{\beta}\, t_{\rm ath}, \qquad v_{\rm ph} = v_{\rm ath}/\sqrt{\beta}, \qquad \rho_{\rm ph} = \rho_{\rm ath}\,\frac{\rho_0 L_0^3}{M_{\rm bh}}, \qquad P_{\rm orb} = 2\pi\sqrt{\frac{R_0^3}{\beta}}\Big|_{\rm ath} = 2\pi R_0^{3/2}\Big|_{\rm ph}. \tag{1}$$

T_0 and μ drop out: they fix t_0, which the torus never sees. With $M_{\rm bh} = 4.5{\times}10^7 M_\odot$, $\chi = 500$: $L_0 = 6.645{\times}10^{15}$ cm, $t_u = 7.01\times10^6$ s, $P_{\rm orb} = 1.40$ yr. The torus is not self-gravitating, so $\rho_0 = 10^{-13}\,\mathrm{g\,cm^{-3}}$ sets only the mass scale.

2 Equilibrium

Steady, axisymmetric, $\mathbf{v} = R\Omega(R)\,\hat{\boldsymbol{\phi}}$, $p = K\rho^{1+1/n}$, $n = 1/(\gamma-1)$, point mass $\Phi = -GM/r$, $r = \sqrt{R^2+z^2}$ (PP84 §2):

$$\frac{1}{\rho}\nabla p = -\nabla\Phi + \Omega^2 R\,\hat{\mathbf{R}}, \qquad \frac{1}{\rho}\nabla p = \nabla\Big[(n+1)\frac{p}{\rho}\Big], \qquad \Omega^2 R = \frac{\ell^2}{R^3} = -\frac{\mathrm{d}}{\mathrm{d}R}\Big(\frac{\ell^2}{2R^2}\Big) \quad (\ell = \text{const}),$$

$$(n+1)\frac{p}{\rho} = \frac{GM}{r} - \frac{\ell^2}{2R^2} - C.$$

$\nabla p = 0$ at $(R_0, 0)$ gives $\ell^2 = GMR_0$; write $C = GMC'/R_0$:

$$\frac{p}{\rho} = \frac{GM}{(n+1)R_0}\,f, \qquad f(R,z) = \frac{R_0}{r} - \frac{1}{2}\Big(\frac{R_0}{R}\Big)^2 - C', \qquad f_c = f(R_0,0) = \tfrac{1}{2} - C'. \tag{2}$$

$p/\rho = K\rho^{1/n} \propto f$, normalised to ρ_c at $(R_0, 0)$:

$$\rho = \rho_c\Big(\frac{f}{f_c}\Big)^n \Theta(f), \qquad u = \frac{p}{(\gamma-1)\rho} = \frac{n\,GM}{(n+1)R_0} f, \qquad v_\phi = \frac{\sqrt{GMR_0}}{R}, \qquad c_s^2 = \gamma\frac{p}{\rho}. \tag{3}$$

Surface $f = 0$. In the mid-plane, with $x = R_0/R$: $x^2 - 2x + 2C' = 0$,

$$R_{\rm in,out} = \frac{R_0}{1 \pm \sqrt{1-2C'}}, \qquad z_s(R) = \sqrt{s^2 - R^2}, \quad s = \Big(\frac{R_0}{2R^2} + \frac{C'}{R_0}\Big)^{-1}, \qquad \frac{c_s}{v_\phi}\Big|_{R_0} = \sqrt{(\gamma-1)f_c}. \tag{4}$$

$z = 0$ gives $f(R)$ of the 2D Athena++ runs: the cylinder there is this torus' mid-plane continued uniformly in z, with gravity $-\beta/R^2$ in R only. Parameters: athena $C' = 0.2$, $\gamma = 1.3$: $R_{\rm in} = 0.5635$, $R_{\rm out} = 4.4365$, $z_{\rm top} = 2.564$, $c_s/v_\phi = 0.300$. PP84 models ($n = 3$, $d = 1/2C'$): $C' = 0.495,\ 0.444,\ 0.25$.

3 SPH (Phantom)

Density, smoothing length and momentum (cubic M_4, $h_{\rm fact} = 1.2$):

$$\rho_a = \sum_b m\, W(|\mathbf{r}_a - \mathbf{r}_b|, h_a), \qquad h_a = h_{\rm fact}\Big(\frac{m}{\rho_a}\Big)^{1/3}, \tag{5}$$

$$\frac{\mathrm{d}\mathbf{v}_a}{\mathrm{d}t} = -\sum_b m\Big[\frac{P_a}{\Omega_a\rho_a^2}\nabla_a W_{ab}(h_a) + \frac{P_b}{\Omega_b\rho_b^2}\nabla_a W_{ab}(h_b)\Big] + \mathbf{a}_a^{\rm AV} - \frac{GM\,\mathbf{r}_a}{r_a^3}, \tag{6}$$

$$\frac{\mathrm{d}u_a}{\mathrm{d}t} = \frac{P_a}{\Omega_a\rho_a^2}\sum_b m\,\mathbf{v}_{ab}\cdot\nabla_a W_{ab}(h_a) + \Lambda_a^{\rm AV}, \qquad P_a = (\gamma-1)\rho_a u_a, \tag{7}$$

$\Omega_a = 1 - \partial h_a/\partial\rho_a \sum_b m\, \partial W_{ab}/\partial h_a$. Shock viscosity with signal speed $v_{\rm sig} = \alpha c_s + \beta|\mathbf{v}_{ab}\cdot\hat{\mathbf{r}}_{ab}|$ and the Cullen–Dehnen switch $\alpha \in [0,1]$, $\alpha_u = 0$ (§5); it is the only dissipation (no physical viscosity), and acts as a shear viscosity of order $\alpha_{\rm SS} \sim 0.1\,\alpha\, h/H$. Particles inside $r_{\rm acc} = 0.35$ (athena's $x_{1,\rm min}$) are accreted. No ambient medium: the surface is free.

4 Particle lattice

Mass $m = \rho_c V/N$, $V = \int (f/f_c)^n \,\mathrm{d}V$. Isotropic spacing $d = (m/\rho)^{1/3}$.

Columns. $\lambda(R) = 1/d(R,0)$, $N_{\rm col} = \text{round}\int_{R_{\rm in}}^{R_{\rm out}} \lambda\,\mathrm{d}R$; column edges at equal steps of $\Lambda(R) = \int_{R_{\rm in}}^{R} \lambda\,\mathrm{d}R'$, centre R_c, width ΔR.

Rings. Half-column mass $M_z(z) = 2\pi R_c \Delta R\,\rho_c \int_0^z (f/f_c)^n \mathrm{d}z'$. A ring at height z holds

$$N_\phi = n_s \max\Big(1, \text{round}\frac{2\pi R_c}{n_s\, d(R_c,z)}\Big), \qquad z_k = M_z^{-1}\big(M_{\rm lo} + \tfrac{1}{2}N_\phi m\big), \qquad M_{\rm lo} \leftarrow M_{\rm lo} + N_\phi m, \tag{8}$$

mirrored to $-z_k$; the mid-plane ring takes $\frac{1}{2}N_\phi m$ from each half. The mass of every ring matches the analytic column, so rounding N_ϕ stretches the ring in z (by at most $n_s/2N_\phi$) instead of misweighting it. $\phi_j = (j - 1 + \frac{1}{2}s)\,2\pi/N_\phi$, $s \in \{0,1\}$ alternating.

Symmetry. For one ring, $\sum_{j=1}^{N_\phi} e^{-im\phi_j} = e^{-im\phi_1}\frac{1-e^{-2\pi i m}}{1-e^{-2\pi i m/N_\phi}} = 0$ unless $N_\phi \,|\, m$. Every N_ϕ is a multiple of n_s, so $A_m = 0$ for $n_s \nmid m$, to round-off (5×10^{-17} measured). With Phantom's $\pi = 3.1415926536$ the ring spans $2\pi(1 + 7\times10^{-12})$ and the open gap gives 3×10^{-12} in every m: the setup uses $\pi = 4\arctan 1$.

5 Relaxation and spin-up

For $\ell = $ const the centrifugal term is a potential force, $\Omega^2 R = \ell^2/R^3 = -\partial_R(\ell^2/2R^2)$, so the rotating torus is the static gas of

$$\Phi_{\rm eff} = -\frac{GM}{r} + \frac{\ell^2}{2R^2}, \qquad -\nabla\Phi_{\rm eff} = -\frac{GM\,\mathbf{r}}{r^3} + \frac{\ell^2}{R^3}\hat{\mathbf{R}}, \qquad \frac{1}{\rho}\nabla p = -\nabla\Phi_{\rm eff} \tag{9}$$

(Phantom's `iexternalforce = 5`, written for $GM = R_0 = \ell = 1$). Stage 1: $\mathbf{v} = 0$, $\mathrm{d}\mathbf{v}_a/\mathrm{d}t = \cdots - \nabla\Phi_{\rm eff} - \gamma_d\mathbf{v}_a$ ($\gamma_d = 0.3$, 2 orbits): the particles settle into the discrete SPH equilibrium, which differs from the analytic one near the surfaces. No shear, so the n_s-fold symmetry survives ($A_m/A_0 \sim 10^{-10}$ for $m < 4$ after 2 orbits). Stage 2: $v_\phi = \ell/R$ on the relaxed positions replaces the effective centrifugal force by the real one, ℓ^2/R^3, so every particle keeps its force balance; then the seed, the point mass, $\gamma_d = 0$, $t = 0$.

No artificial conductivity ($\alpha_u = 0$). Its signal speed $v^u_{\rm sig} = (|P_a - P_b|/\bar\rho_{ab})^{1/2}$ is non-zero between neighbours of any stratified equilibrium, $|P_a - P_b| \sim \rho g h$, so it carries heat down the gradient of $u \propto f$: relaxing at rest with $\alpha_u = 1$, $\rho_{\max}$ rose 32% in 6 orbits and the envelope blew past $R_{\rm out}$; with $\alpha_u = 0$, $E_{\rm kin}/E_{\rm th} \to 10^{-5}$ and $\rho_{\max}$ holds to 0.3%. The torus is homentropic: there is no contact to smooth.

6 Seed (as athena-vv)

$$\delta v_R = A\, c_s\, \frac{f}{f_c}\, S(\phi), \qquad S = \begin{cases} \sin m\phi & \text{single} \\ m^{-1/2}\sum_{k=1}^{m}\sin(k\phi + \pi\xi_k) & \text{multi} \\ \xi_a \in [-1,1)\ \text{per particle} & \text{noise} \end{cases} \tag{10}$$

$\delta\rho = \delta p = 0$; the envelope f/f_c vanishes at the surface.

7 Diagnostics

Annuli k of width ΔR on $[R_{\rm lo}, R_{\rm hi}]$, cloud-in-cell weights w_{ak}:

$$\Sigma_m(R_k) = \frac{m}{2\pi R_k \Delta R}\sum_a w_{ak}\, e^{-im\phi_a}, \qquad A_m = \Big[\sum_k |\Sigma_m(R_k)|^2 R_k \Delta R\Big]^{1/2}, \tag{11}$$

Fourier first, radius second (radial lobes cannot cancel); F_m the same with weights $w_{ak}v_{R,a}$. Linear regime: $A_m/A_0 < 0.02$. Pattern speed $-\dot\psi_m/m$, $\psi_m = \arg\Sigma_m(R_0)$. Shot noise of a disordered set: $A_m/A_0 \sim (n_{\rm bin}/N)^{1/2}$.

Maps (scatter interpolation): $\rho(\mathbf{r}) = \sum_b mW(|\mathbf{r} - \mathbf{r}_b|, h_b)$, $\langle q\rangle = \sum_b (m/\rho_b)q_b W_b / \sum_b (m/\rho_b)W_b$; azimuthal mean $\langle\rho\rangle_\phi(R,z) = \sum_b \frac{m}{2\pi R} W_{2D}(|(R,z) - (R_b,z_b)|, h_b)$ (ring projection, $h \ll R$).

8 Linear theory (PP84 §3–4)

Linearise about $\mathbf{v}_0 = R\Omega\hat{\boldsymbol{\phi}}$, perturbations $\propto e^{i(m\phi+\sigma t)}$, $\bar\sigma = \sigma + m\Omega$, $p'/p = \gamma\rho'/\rho$, $W = p'/(\rho\bar\sigma)$. For $\ell = \text{const}$ the epicyclic frequency $\kappa^2 = R^{-3}\mathrm{d}(R^4\Omega^2)/\mathrm{d}R = 0$, and

$$\frac{1}{R}\frac{\partial}{\partial R}\Big(\rho R\frac{\partial W}{\partial R}\Big) + \frac{\partial}{\partial z}\Big(\rho\frac{\partial W}{\partial z}\Big) - \frac{m^2}{R^2}\rho W = -\bar\sigma^2\frac{\rho^2}{\gamma p}W, \tag{12}$$

i.e. $\sigma^2\mathsf{A}W + \sigma\mathsf{B}W + \mathsf{C}W = 0$ with $\mathsf{A} = \rho^2/\gamma p$, $\mathsf{B} = 2m\Omega\mathsf{A}$, $\mathsf{C} = \mathcal{L} + m^2\Omega^2\mathsf{A}$. Linearised:

$$\begin{pmatrix} -\mathsf{B} & -\mathsf{C} \\ \mathsf{I} & 0 \end{pmatrix} x = \sigma \begin{pmatrix} \mathsf{A} & 0 \\ 0 & \mathsf{I} \end{pmatrix} x, \qquad x = (\sigma W, W), \tag{13}$$

growth $-\mathrm{Im}\,\sigma$, pattern $-\mathrm{Re}\,\sigma/m$, e-folds per orbit $2\pi(-\mathrm{Im}\,\sigma)/\Omega_0$. Conservative cell-centred (R, z) grid, $z \geq 0$ with $\partial_z W = 0$ at $z = 0$ (even modes); faces into cells with $\rho = 0$ carry no flux, which is PP84's regularity condition at the surface. Dropping ∂_z gives the 2D cylinder.

model, m	growth time $[\Omega_0^{-1}]$ this solver	PP84 A / B
1 ($C' = 0.495$), 2	8.48	8.0 / 8.5
1, 4	4.70	3.9 / 4.7
1, 6	3.88	2.9 / 4.0
2 ($C' = 0.444$), 2	4.57	4.0 / 4.6
3 ($C' = 0.25$), 2	9.32*	8.5 / 10.7

athena torus m	e-folds / orbit 3D torus	2D cylinder
1	0.798	1.086
2	0.662	0.976
3	–	0.759
4	–	0.587

* 40×20 grid, 10% from 20×10. 3D: 160×80, converged to $< 0.3\%$; $m \geq 3$ in 3D not yet converged. 2D: 400 cells, identical to athena-vv's `ppi.py` at $\gamma = 4/3$.